COMP9334: Capacity Planning of Computer Systems and Networks

Course Review and Exam Info

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System performance is important

- Performance metrics: response time, waiting time, throughput, availability
- Performance is determined by:
  - Workload
  - System parameters
- You can estimate system performance, without building the actual system, by using queueing models

Performance analysis techniques (1)

- Operational analysis (1B,2A)
  - Measurements on the systems
  - Operational laws, in particular Little's Law
  - Key concept: Bottleneck
  - Upper bound on system performance

Performance analysis techniques (2)

- Need to identify the inter-arrival and service time distributions

- Queues with Poisson arrival
  - Exponential service time (2B,3A)
    - Single or multiple servers: M/M/1 versus M/M/m
    - Infinite buffer or finite buffer: M/M/m versus M/M/m/m+k

  - General service time distribution (4A,7A)
    - M/G/1. Key concept: residual service time
    - Priority queueing

Performance analysis techniques (3)

- Closed queueing networks with exponential service time
  - Markov chain analysis (3B)
    - Recipe: Identify states and transition rates; set up balance equations; solve steady state probabilities; determine performance
  - Mean value analysis (7B)
    - Iterative method
    - $n = 0$ jobs $\rightarrow$ $n = 1$ job $\rightarrow$ $n = 2$ jobs $\rightarrow$ ...

Performance analysis techniques: some key points

- No universal analytical methods
  - Analytical solutions are only available for specific classes of queues
  - Upper bounds are only available for some general classes of queues
- Simulation can be used to determine general queueing problems

Simulation (1)

- How to do discrete event simulation?

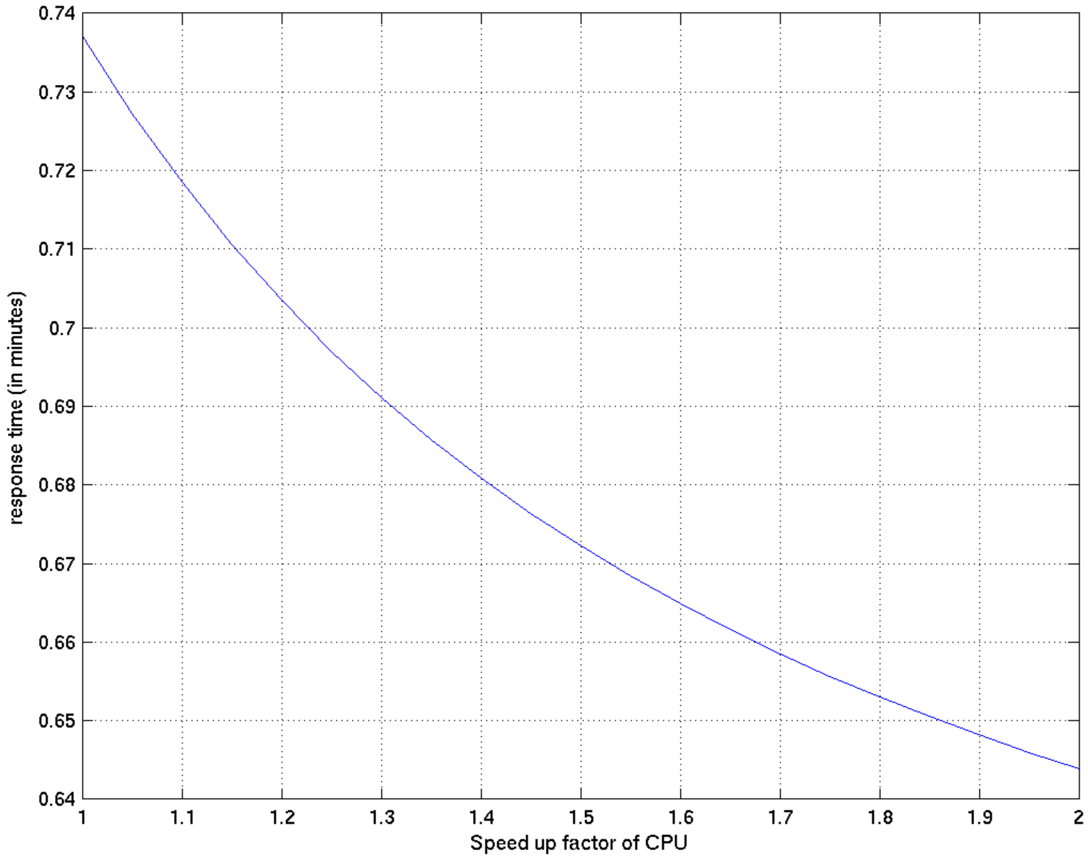
- How to generate random events according to the specific inter-arrival time and service time distributions
  - Generating uniformly distributed pseudo-random numbers
  - The inverse transform method

Simulation (2)

- Simulation is not just about writing correct simulation code (though it is important), it is also important to do sound statistical analysis on the simulation results obtained
- Transient removal
- Independent replications
- Confidence interval
- How to decide whether one system is better than the other using confidence interval?
  - Paired observations: Paired-t confidence interval
  - Approximate visual test
- Variance reduction method

Capacity planning and performance analysis

- Solve the capacity planning problem by solving a number of performance analysis problems
- Example: Revision Problem: Week 3B, Question2



Applications of queueing (7A_2,8A)

- Web services
  - Fork-join queues
- Other applications
  - Determining a good multi-programming level
  - Power allocation in server farm
  - Server farm with set-up cost

Integer programming (1)

- Linear programming (LP)
  - Real values for decision variables, linear in objective function, linear in constraints
- Integer programming (IP)
  - Some decision variables can only take integer values
  - Some decision variables can only take binary values, e.g. for making yes-or-no decisions

Integer programming (2)

- Applications of integer programming in network flow problems
  - Flow conservation constraints to ensure a unique path between two nodes in the network
- Example applications
  - Traffic engineering
  - Network design

Integer programming (3)

- Applications of integer programming in placement problems
  - Placement of wireless access points
  - Placement of controllers in software-defined networks (revision problem)
- Power of binary variables
  - Restricted range of values
  - Either-or constraints
  - Piecewise linear functions
- Important to make the constraints linear

Summary

- What you have learned through this course are fundamental techniques that can be applied to design computer systems and networks to have good performance
  - We hope you have gained some skills from this course
  - We hope you have been trained to be performance-minded
- Due to the limited time and scope of this course, we cannot cover all techniques that have been developed in this field
- However, with the knowledge you have acquired from this course, you should have the foundation to learn more ...
- Mathematical methods can be applied to many different areas

Final exam

- Tue 9 May 2023, 1:45-4pm (Sydney time)
  - 2 hours and 15 minutes
  - If you are not in Sydney, make sure you know your corresponding local time

- Exam syllabus:
  - Not examined:
    - Weeks 4B, 5A, 5B_1, 5B_2: Discrete event simulation
    - Week 7A_2 which is a lecture based on technical papers
  - No questions on programming or AMPL
  - No extensive computation
  - All other topics are examinable

Final exam (cont.)

- UNSW student code, which covers exam conducts, applies
- This linked page covers the following aspects of the exam. (Note: The page is also available at the course website under Week 10's lecture.)
  - Mode of attendance
  - Invigilation
  - Inspera (Device and OS requirements. How to prepare your device for the exam.)
  - What you must bring, can bring and cannot bring to the exam
  - Open book

Answering questions in the final exam

- Question aims to test understanding, not memorization
  - Style similar to revision problems, assignment and sample exam
- Show your steps in your answer
  - If you only write the final answer, you won't receive full marks even if it is correct
  - You will receive marks for your steps
  - You can use the solution to revision problems as a guide as to how much explanation you need

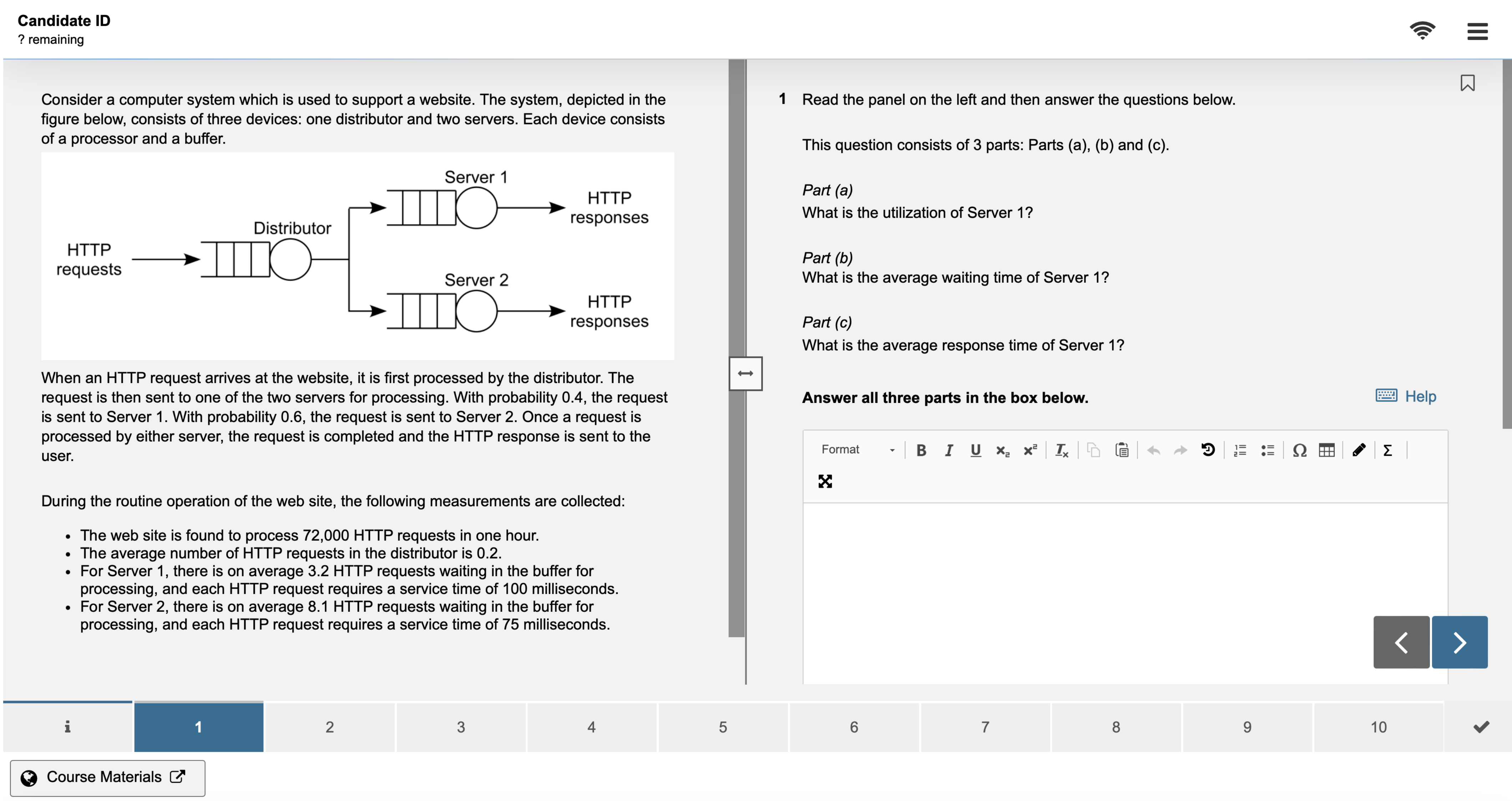
Inspera - a browser based exam platform

- The exam will use Inspera (screen shot on next slide)

- Should try to familiarise yourselves with the platform
  - This linked page, which is also mentioned on Page 15, points to UNSW web pages on Inspera.
  - Practice using the sample exam

- Answer is auto-saved. Your answers will be automatically submitted at the end of the exam unless your Internet connection is lost.

Screen shot of Inspera

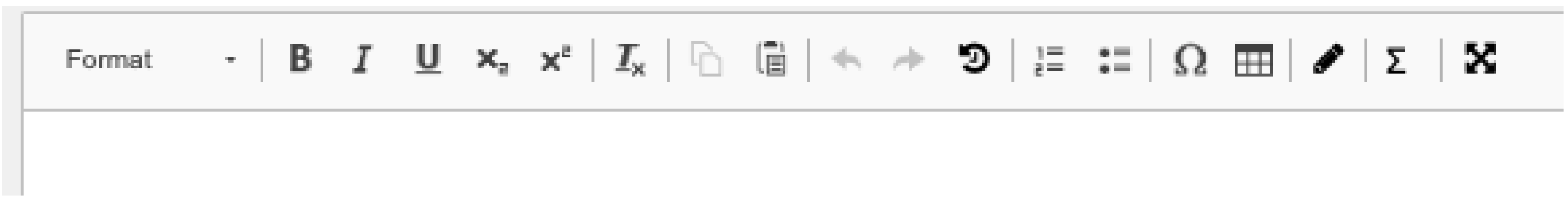


Answer box and answer formatting

- For COMP9334, all questions use the same type of Inspera answer box.

- This applies to the sample exam and the actual exam.

- I am more interested to know whether you understand the subject matter or not
  - Write and format in a way to help me to understand your idea
  - Although Inspera has a good editor, some maths formatting is optional, e.g. OK to write `p_{123}` instead of p_{123}, `10^6` instead of 10^6, `lambda` instead of λ etc.



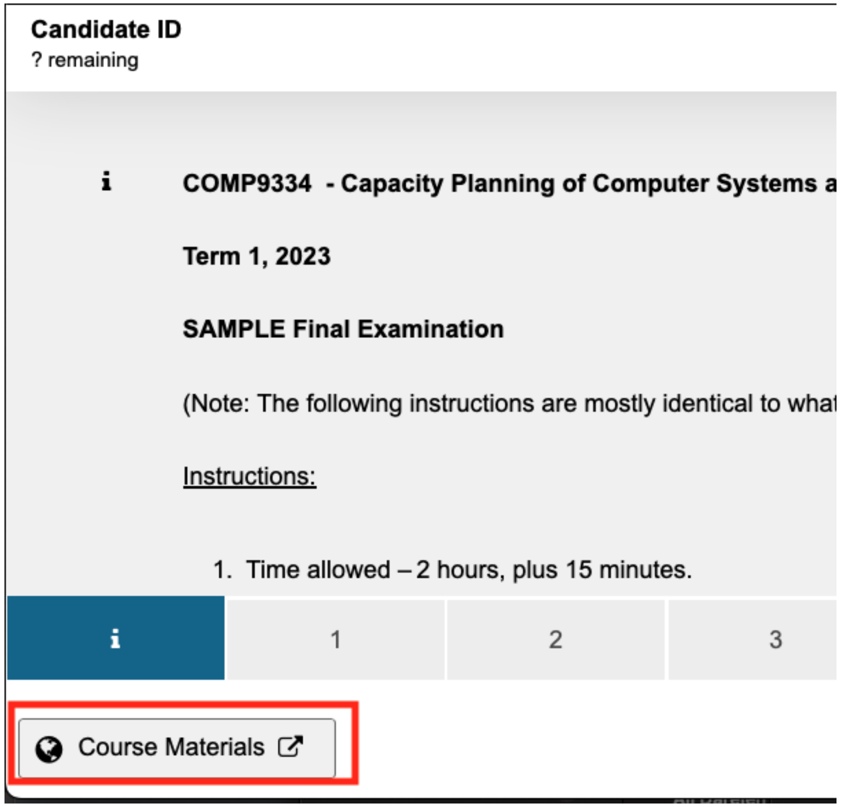
Preparations for final exam

- Make sure you understand all the concepts, techniques and examples discussed in the lectures
- Work through all the sample problems, assignment questions, etc for practice
- Sample exam on Moodle (more on this later)
- Misconception: Open-book exam means no preparation required
- Consultations: See course website under Timetable
- Further questions
  - Post on Forum (Try to avoid last minute questions.)

Sample exam on Moodle

- Accessible via the COMP9334 Moodle page
- Most questions are based on past exams
- Settings
  - Available from 4pm on 20 April 2023
  - You can access it multiple times
  - No time limit set - you can time yourselves if you wish
- Do **not** submit because you will not be able to access it again after you have submitted
- A sample solution will be provided

- The cover page of the sample exam is almost the same as what you will see in the exam
  - This is so that you know the instructions in advance
- Note the ”Course Materials” link is at the bottom left hand corner, see below. (Same for final exam.)



Exam policy

- Exam policy
  - If you attend the exam, we assume that you are functioning well on that day and will not offer you a supplementary exam.
  - In case you are not well on the day of the exam, get a medical certificate and do not attend the exam.
  - If you fall ill during the exam, you should
    - Stop working on the exam.
    - Notify the invigilator and ask for instructions.
    - Get a doctor's certificate and submit a special consideration.

Parting messages

- Please complete the myExperience survey
  - Good/bad/more of this/less of that/what can be done better
- This course is different from many CSE courses
- Analytical and simulation methods are useful for many disciplines
- This world needs people with multiple skills (hard and soft). Important to find your talents and passions, but try to explore and learn as many different areas as you can.